Exam in Numerical Analysis E3, FMN050, 040819

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This exam lasts 14:00–17:00. In order to pass, a minimum of 15 points out of the total 30 is required. The final grade of the course will be: for **grade 3**, 15–19 points are required; **grade 4**: 20–24 p, **grade 5**: 25–30 p (preliminary figures).

During the exam you are allowed to use a pocket calculator and the course lecture notes (copies of slides). No other material is allowed. Please answer the questions in Swedish or English.

Problems

A. Condition and Error analysis

1. Let $F(x_1, x_2, x_3)$ be a linear, scalar function of the three scalar variables x_1, x_2 and x_3. Let x denote the vector with these three components. The function can then be written $F(x) = c^{\mathrm{T}}x$ for some vector c. Let F be evaluated at x and at the perturbed point $x+\delta x$. Under what conditions on the perturbation could it happen that $F + \delta F = F(x + \delta x) = F(x)$? **(1p)**

Solution. We have $F(x) = c^{\mathrm{T}}x$. Then $F'(x) = \operatorname{grad} F = c^{\mathrm{T}}$. So, $\delta F = F'(x)\delta x = c^{\mathrm{T}}\delta x$. This is zero if c and δx are orthogonal.

2. Now assume that the function F is nonlinear instead. It is again calculated at x and at $x + \delta x$. Can it still happen that $F + \delta F = F$? If so, state the condition on F and δx causing this to happen. (Neglect higher order terms.) **(1p)**

Solution. Now $\delta F \approx F'(x)\delta x = \operatorname{grad} F \cdot \delta x$. So if δx is orthogonal to the gradient of F, then $F + \delta F \approx F$ when higher order terms are neglected.

3. Consider the special case $F(a, b, c) = (ab + cb)/(abc)$ and give the corresponding conditions on δa, δb and δc. **(2p)**

Solution. Here $F = 1/c + 1/a$ is independent of b, so δb can have any value without affecting F. As

$$F'(x)\mathrm{d}x = -\mathrm{d}c/c^2 - \mathrm{d}a/a^2$$

we see that $\delta F \approx 0$ if $\delta c/\delta a = -c^2/a^2$.

B. Interpolation and Least squares

1. Fit a quadratic polynomial by the least squares method to the data in Table 1. **(3p)**

Solution. We work with a shifted polynomial

$$P(x) = c_0 + c_1(x - 10) + c_2(x - 10)^2$$

x	8	10	11	12
y	100	101	101	102

Table 1: Data for problem **B**1.

and fit it to $y = 100 + z$. Thus we want to solve

$$\begin{bmatrix} 1 & -2 & 4 \\ 1 & 0 & -0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

We get the normal equations

$$\begin{bmatrix} 4 & 1 & 9 \\ 1 & 9 & 1 \\ 9 & 1 & 33 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix},$$

with the solution $c_0 = 86/110$, $c_1 = 51/110$ and $c_2 = 5/110$. The polynomial is then

$$P(x) = 100 + 86/110 + 51(x-10)/110 + 5(x-10)^2/110,$$

or

$$P(x) \approx 100.7818 + 0.4636(x-10) + 0.0455(x-10)^2.$$

2. a) The Dow Jones industrial index in 2001 is given in Table 2. Find the polynomial of the lowest possible degree interpolating the data from March to May by applying Lagrange interpolation. **(2p)**

Solution. Interpolation the three points for March, April and May implies that we need a 2nd degree polynomial. The basis functions are

$$\phi_0 = \frac{(x-4)(x-5)}{2}; \quad \phi_1 = \frac{(x-3)(x-5)}{-1}; \quad \phi_2 = \frac{(x-3)(x-4)}{2}$$

The interpolation polynomial is then

$$P(x) = 10400\,\frac{(x-4)(x-5)}{2} + 9800\,\frac{(x-3)(x-5)}{-1} + 10800\,\frac{(x-3)(x-4)}{2}.$$

b) Evaluate the polynomial to estimate the index in October. Conclusion? **(1p)**

Solution. We evaluate $P(10)$ and find

$$P(10) = 10400\,\frac{6\cdot 5}{2} + 9800\,\frac{7\cdot 5}{-1} + 10800\,\frac{7\cdot 6}{2} = 39800.$$

Conclusion? The interpolated (in fact, extrapolated) index is way off the mark. Using such extrapolation is unwarranted.

month	1	2	3	4	5
index	10600	11000	10400	9800	10800

month	6	7	8	9	10
index	11000	10500	10500	10000	8800

Table 2: Data for problem **B**2.

C. Integration

1. In a homework assignment you found an approximation to the average temperature of a brake system. The average temperature is given by

$$T_{avg} = \frac{\int_{9.38}^{14.58} T(r) r dr}{\int_{9.38}^{14.58} r dr},$$

where $T(r)$ is the temperature at distance r from the brake centre. Approximate T_{avg} from the reduced set of values given in Table 3 by applying the trapezoidal rule. **(4p)**

r	9.38	10.42	11.46	12.50	13.54	14.58
$T(r)$	338	474	557	601	651	671

Table 3: Temperature data for problem **C**1.

Solution. We compute the two integrals using the trapezoidal rule.

$$\int_{9.38}^{14.58} T(r) r dr \approx 1.04 \cdot \left(\frac{9.38 \cdot 338}{2} + 10.42 \cdot 474 + \ldots + 13.54 \cdot 651 + \frac{14.58 \cdot 671}{2}\right)$$

which gives the result 35491. Similarly, the second integral is

$$\int_{9.38}^{14.58} r dr \approx 1.04 \cdot \left(\frac{9.38}{2} + 10.42 + \ldots + 13.54 + \frac{14.58}{2}\right) \approx 62.296.$$

Hence the average value is $35491/62.296 \approx 569.7$ degrees.

D. Newton's method

1. We are going to solve the the nonlinear system

$$\begin{aligned} x^2 - \sin xy &= 0 \\ 4x \sin y - x &= 1 \end{aligned}$$

using Newton's method, with $(x^{(0)}, y^{(0)}) = (1, 0)$ as the initial guess.

a) Construct the Jacobian matrix at the starting point. **(1p)**

Solution.

$$\begin{bmatrix} 2x - y\cos xy & -x\cos xy \\ 4\sin y - 1 & 4x\cos y \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix}.$$

b) Introduce a suitable notation and state the algorithmic steps for the Newton iteration. **(1p)**

Solution The algorithm consists of the steps

1. Compute residual $f(x^{(k)}, y^{(k)})$
2. Compute Jacobian $f'(x^{(k)}, y^{(k)})$
3. Solve the linear system $f'(x^{(k)}, y^{(k)}) \cdot \delta z = -f(x^{(k)}, y^{(k)})$
4. Update $(x^{(k+1)}, y^{(k+1)}) = (x^{(k)}, y^{(k)}) + \delta z$

c) Perform one iteration of Newton's method from the starting point and compute the approximation $(x^{(1)}, y^{(1)})$. **(2p)**

Solution At the starting point $(1, 0)$ we have $f = (1, -2)$. Taking one step with the algorithm above then produces $(x^{(1)}, y^{(1)}) = (5/7, 3/7) \approx (0.7143, 0.4286)$.

E. ODE Initial Value Problem

1. Determine $y(0.4)$ and $z(0.4)$ for

$$\begin{aligned} \mathrm{d}y/\mathrm{d}t &= 1 - tyz \\ \mathrm{d}z/\mathrm{d}t &= 1 - z - y \end{aligned}$$

with initial data $y(0) = 1$ and $z(0) = 1$, and using the explicit Euler method with step size $h = 0.2$. **(4p)**

Solution At the starting point the right hand side is $(1, -1)$. The first step then produces

$$y_1 = 1 + 0.2 \cdot 1 = 1.2$$

and

$$z_1 = 1 + 0.2 \cdot (-1) = 0.8.$$

At this new point, the right hand side is $(1 - 0.2 \cdot 1.2 \cdot 0.8, 1 - 0.8 - 1.2) = (0.808, -1)$ (note that $t = 0.2$). Repeating the step gives

$$y_2 = 1.2 + 0.2 \cdot 0.808 = 1.3616$$

and

$$z_2 = 0.8 + 0.2 \cdot (-1) = 0.6.$$

The solution at $t = 0.4$ is therefore approximately $(1.36, 0.6)$.

F. Linear systems

1. When solving linear systems $Ax = b$ with iterative methods, the matrix A is split into $A = L+D+U$. The Jacobi method iterates according to

$$Dx^{(k+1)} = -(L+U)x^{(k)} + b$$

Use this method, starting from $x^{(0)} = 0$, to perform one iteration on the system

$$\begin{bmatrix} 5 & -2 & 0 \\ 1 & 4 & -2 \\ 4 & -3 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}.$$

(3p)

Solution Here we have

$$D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

and

$$L+U = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & -2 \\ 4 & -3 & 0 \end{bmatrix}.$$

Taking one iteration from 0 means that we have to solve

$$D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix} x^{(1)} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

so the solution is $(1/5, -1/2, 1/3)$.

G. Quick questions (True or false; 0.5p for each correct answer).

1. A problem is ill–conditioned if its solution is highly sensitive to small changes in the problem data. **TRUE**

2. When constructing an interpolation polynomial, the condition number of the (coefficients of the) interpolation polynomial is independent of the choice of polynomial basis (e.g. monomial basis, Lagrange interpolation, Newton interpolation etc.). **FALSE**

3. The optimal way of fitting a straight line to given data is independent of what norm is used to measure the residual. **FALSE**

4. When a numerical time-stepping method is used to solve the initial value problem $\dot{y} = \lambda y$ with $\lambda \in \mathbb{C}$, then stability depends only on $\lambda \cdot \Delta t$, where Δt is the time step. **TRUE**

5. The reason one computes the LU-factorization of a matrix A when solving a linear system $Ax = b$ is that it is much more efficient than inverting the matrix A. **TRUE**

6. Iterative methods like Jacobi or Gauss-Seidel are less expensive than LU factorization if the matrix is large and very sparse ("gles matris"). **TRUE**

7. In the fast Fourier transform (FFT) the matrix–vector multiplication $c = Fy$ can be carried out in only $n \log_2 n$ operations when the dimension of F is a power of 2. If one knows c and wants to compute y (i.e., one wants to solve $Fy = c$), how many operations are needed? **ANSWER:** $n \log_2 n$

8. If one solves a nonlinear equation $x = g(x)$ by a convergent fixed point iteration, then the convergence rate is usually quadratic. **FALSE**

9. If $\dot{y} = f(y)$ is solved using the implicit Euler method then one must solve a linear equation of the form $y = hf(y) + \psi$ on each time step. **TRUE**

10. The order of a quadrature rule for $\int_a^b f(x)\,\mathrm{d}x$ is p if the numerical result is exact for all polynomials of degree $p - 1$ or lower. **TRUE**

(5p total)